



Impacts of lithium tetrafluoroborate and lithium difluoro(oxalate)borate as additives on the storage life of Li-ion battery at elevated temperature

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Abstract

The impacts of boron-based Li salt additives including lithium tetrafluoroborate (LiBF_4) and lithium difluoro(oxalate)borate (LDFOB) on the storage life of Li-ion battery at elevated temperature are investigated. Adding 1 wt% additives in the electrolyte significantly affects the storage life of the $\text{LiNi}_{0.8}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2$ /graphite full cell at 55 °C. The anode solid electrolyte interphase (SEI), preventing the loss of Li^+ and e^- in anode, is the key factor affecting the storage life. The formation and aging of SEI on the graphite anode with and without additives are investigated. It is found that the SEI formed with the addition of LiBF_4 is thick and loose due to LiF crystals produced by the decomposition of LiBF_4 and the SEI cannot prevent the Li^+ and e^- loss in anode and the decomposition of the electrolyte solvent, resulting in shorter storage life of the battery. On the contrary, the SEI formed with the addition of LDFOB is thick and compact due to formation of the lithium oxalate in the SEI, produced by the decomposition of LDFOB. The SEI efficiently inhibits decomposition of the electrolyte solvent on anode and makes a longer storage life of the battery.

Keywords Li-ion batteries · Additives · Storage life · Elevated temperature

Introduction

Recently, commercial lithium-ion batteries (LIBs) have performed well for most electronic applications. However, currently available LIB technology still do not satisfy some of the performance goals for hybrid electric vehicles (HEV) or plug-in hybrid electric vehicles (PHEV). The restrained lifetime is one of the significant problems with current LIBs, especially at elevated temperature while the aging progress would be accelerated [1, 2]. The lifetime of LIBs is expected to be as long as 10–15 years in automotive applications [3]. Hence, much more work still needs to be done to extend the life performance of current LIBs.

The lifetime performance for LIB application includes the storage life and cycle life. However, the LIBs are complex systems and the processes of their aging are even more complicated in application while the storage aging and cycle aging always occur in combination, the capacity decrease and power fading during aging do not originate from one single cause but originate from a number of various processes and their interactions [4–6]. Hence, to get a better understanding and assessment of the aging mechanisms and lifespan, the two life performances are often discussed separately [7, 8]. The storage life is our main concern in this work.

The main cause of the capacity loss during aging is thought to be the degradation reactions of electrolyte with lithium in anode, resulting in Li^+ and e^- loss in anode, along with the SEI sustained growing and becoming more efficient in preventing Li^+ and e^- loss and decomposition of electrolyte on graphite anode [4, 6, 9]. While in cathode, the main causes inducing capacity loss are the cathode structure changing, mainly the NiO layer generation in lithium nickel oxide-based cathode [10, 11]. However, the structure changing was significant during the cycle aging, while during storage aging, no significant changes of surface crystal and electronic structures of $\text{LiNi}_{0.8}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2$ cathode could be observed [12–14]. Hence, the anode SEI was our main concern during storage

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aging. The anode SEI is the protective layer that inhibits the decomposition of electrolyte on anode [15]; therefore, a more stable and better formed SEI would restrain the storage aging of LIBs. The introduction of novel SEI former additives is an effective way at present [16].

Boron-based Li salts including LiBF_4 and LDFOB have been reported for forming more thermal stability SEI on anode and improving the cycle performance of LIBs [17–20]. Hence, in this study, we examined the impacts of LiBF_4 and LDFOB as salt type additives on the storage life at elevated temperature in full cell at 100% state of charge with $\text{LiNi}_{0.8}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2$ as cathode and graphite as anode. The anode SEI morphology and composition before and after storage aging were investigated to explain the impacts on the storage life.

Experiment

Cell design

$\text{LiNi}_{0.8}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2$ /graphite full cells (polymer-coated aluminum vacuum-sealed bag cells) with capacity 980 ± 20 mAh were assembled. Table 1 lists the components of the electrodes and electrolytes used and their specifications. Before electrolyte was added, the assembled cells were dried at 70°C under vacuum for 24 h to remove any trace water. Electrolyte was added, and cells were sealed in argon-filled glove box under inert atmosphere with $\text{H}_2\text{O} < 5$ ppm and $\text{O}_2 < 10$ ppm. Sealed cells were laid in bake oven for 24 h at 45°C to make the electrodes and separator fully immersed with electrolyte, and then cells were vacuum sealed for electrochemistry test.

Table 1 Bag cell design descriptions

Anode with Cu collector	
Graphite (BTR. Co., Ltd., China)	93%
Carbon black	2%
LA133 binder	5%
Area density	$\approx 110 \text{ g m}^{-2}$
Cathode with Al collector	
$\text{LiNi}_{0.8}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2$ (BTR. Co., Ltd., China)	93%
Carbon black	2%
PVDF binder	5%
Area Density	$\approx 220 \text{ g m}^{-2}$
Base electrolyte (Xiaoyuan. Co., China)	1.2 M LiPF_6 EC: EMC 3:7 (vol%)
Separator	Cellgard 2320
Additives (J&K. Co., Ltd., China)	LiBF_4 , LDFOB

Half cells with pure graphite as positive electrode and lithium foil as negative electrode were prepared to get a better observation of SEI on graphite. Graphite with de-ionized water (1:10 mass ratio) was stirred for 2 h to make the graphite particle dispersed in water. A drop of the lower part of the mixture was carefully applied on the Cu collector (diameter of 15 mm) with a rough surface and then vacuum dried at 80°C for 24 h. Then, 2016 type coin cell was assembled in Ar-filled glove box with lithium foil as negative electrode and 1.2 M LiPF_6 EC:EMC 3:7(vol%) with and without additives as electrolyte.

Three-electrode cell was assembled to analyze the AC impedance of anode extracted from the full cell. Cylindrical discs with a diameter of 6 mm were cut from the anode and used as the working electrode. The lithium foil served as the counter electrode, and the lithium slip was used as reference electrode. 1.2 M LiPF_6 EC:EMC 3:7 (vol%) was served as the electrolyte for all the three-electrode cells.

Electrochemistry test

The coin cells with pure graphite as positive electrode were cycled at C/20 between 0.025–1.5 V for three times at 25°C by Land battery system (Wuhan, China) and then disassembled for TEM observation. The full cells, before storage experiment started, were cycled three times by Qingtian battery system (Guangzhou, China). Then all cells were storage aged at open circuit at 55°C after charged to 4.1 V. For each sampling time during the storage experiment, three cells were first discharged and then cycled to acquire the reversible capacity retention. All the charge and discharge tests for full cells were performed under the rate of C/5 (200 mA) between 2.75 and 4.1 V at 25°C . The AC impedance was observed at a frequency range from 100 kHz to 0.01 Hz with the amplitude of 10 mV, performed in Metrohm Autolab system.

Surface characterization

The pure graphite electrode coin cells, after cycled for three times, were disassembled in Ar-filled glove box. Dimethyl carbonate (DMC) was first used to wash out the graphite on electrode and then allowed dispersion of graphite particle in the DMC solution. The dispersed solution was cast on the copper TEM grid and vacuum dried for 12 h. TEM observation was carried out by Tecnai F20 TEM at 200 kV.

The bag cells, after storage aged and discharged, were disassembled carefully in glove box under Ar atmosphere. The slices of anode were washed by DMC twice to remove trace Li salts and dried under vacuum at room

temperature. The surface morphology of the electrodes was observed with a field emission scanning electron microscope (Hitachi S-4800). Ex situ X-ray photoelectron spectroscopy (Thermo Fisher K-Alpha 1063) were performed using Al K α ($h\nu = 1486.6$ eV) radiation under ultra-high vacuum. TEM and STEM were carried out by Tecnai F20 TEM at 200 kV. For TEM and STEM of aged graphite, graphite particles were extracted from the aged anode and dispersed in DMC solution. The dispersed solution was cast on the copper TEM grid and vacuum dried for 12 h before TEM and STEM observation. Unfortunately, the slice of anode samples were sealed in polymer-coated aluminum bags and transferred to the SEM, TEM, or XPS chambers with exposed to the air about 30 s. The XPS spectra were calibrated by the hydrocarbon peak at the binding energy of 284.5 eV. The spectrum of B1s was not monitored due to the facts that its peak was too close to P1s and low inherent sensitivity. FTIR-ATR analysis of the anode electrodes was carried out with a Bruker V70 spectrometer, with germanium crystal, 4 cm⁻¹ resolution, and 128 scans.

Results and discussion

Storage performance of cells with and without additives

Bag cells were constructed containing base electrolyte (BE) and BE with 1 wt% different additives. The first discharge curves of LiNi_{0.8}Co_{0.15}Al_{0.05}O₂/graphite bag cells after storage aging at 55 °C for 0, 30, 60, 90, and 150 days were depicted in Fig. 1. As shown in Fig. 1, the addition of additives did not have significant effects on the discharge capacity before storage. Meanwhile, all cells suffered from capacity fading during storage aging at elevated temperature. The capacity retention after storage for 30 days at 55 °C was about 88, 80, and 91%, for BE, 1 wt% LiBF₄, and 1 wt% LDFOB electrolytes, respectively. While after 150 days, the capacity retention was about 75, 18, and 80%, respectively. This suggested that the storage aging stability was significantly decreased by the addition of LiBF₄, while the addition of LDFOB exhibited an effective improvement in the

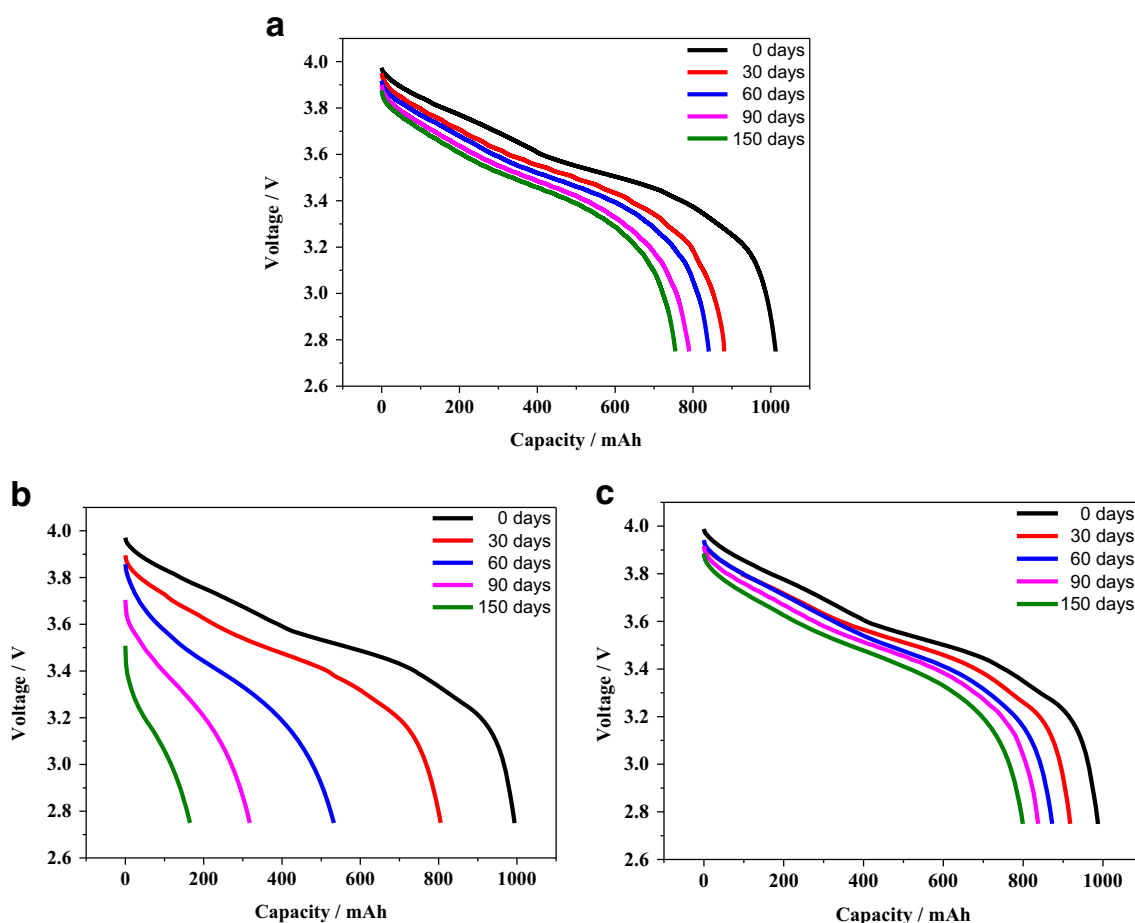


Fig. 1 The discharge curves of full cell after storage at 55 °C for different days with **a** base electrolyte, **b** 1 wt% LiBF₄ electrolyte, and **c** 1 wt% LDFOB electrolyte

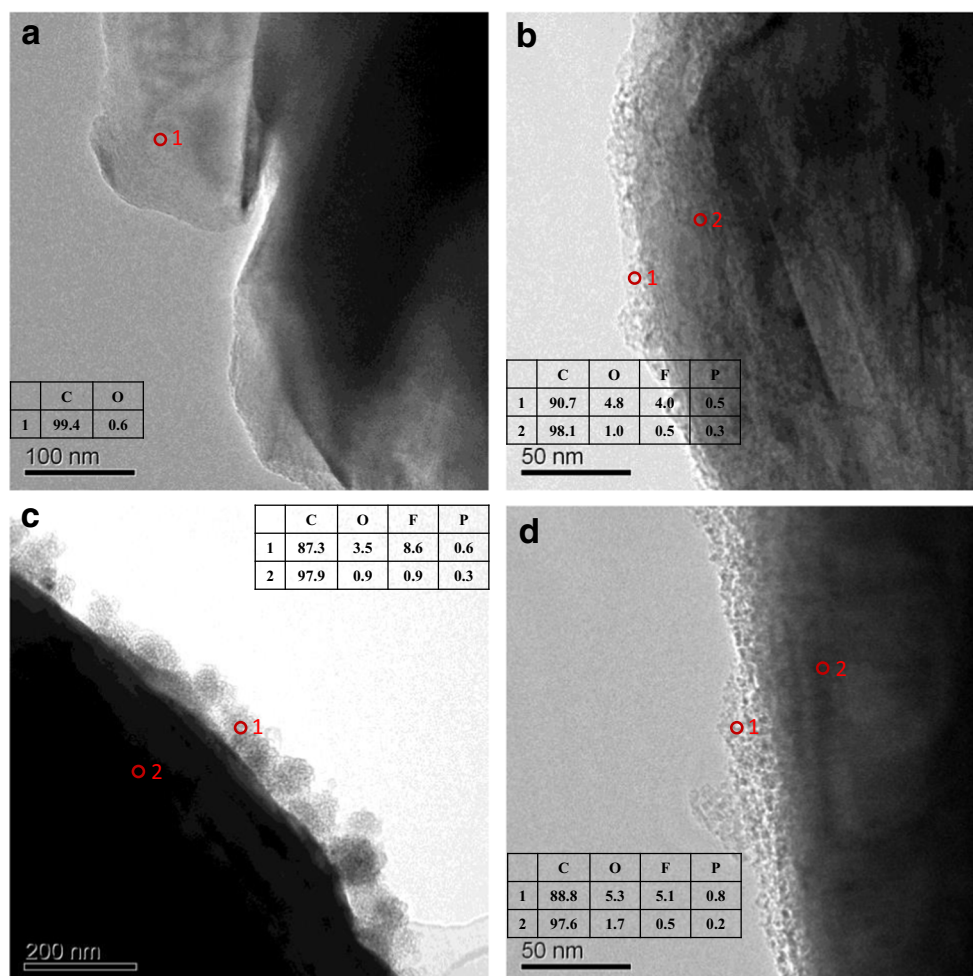
capacity retention during storage aging. Meanwhile, since in most cases it meant the end of life when the capacity loss is larger than 20% of its initial capacity [3], the life span of the cell at 55 °C was about 90, 30, and 150 days, respectively, for base electrolyte, 1 wt% LiBF₄, and 1 wt% LDFOB electrolytes.

The graphite SEI formation with and without additives

The initial (before storage) SEI characteristics, such as thickness, would have notable impacts on the capacity fading based on the reported cell life prediction models [21–23]. To investigate the SEI formed on graphite, pure graphite with Cu collector anode was prepared to analyze the SEI thickness and components through TEM and EDX. The binder was not used since it may be mistaken as SEI under TEM. It was reported that SEI formation was apparent on the edge of graphite after the first lithiation process and became stable in the following cycles [24, 25]. Figure 2 shows the TEM images of graphite which extracted from anodes after 3 cycles

with and without additives when stable SEI was formed. As shown in Fig. 2, the pristine graphite exhibited smooth edges (Fig. 2a) and the composition was dominated by C with a very low content of O. While after cycled, the SEI emerged on the edges of graphite and its morphology and element composition were significantly affected by the addition of additives. The thickness of SEI was about 10, 100, and 20 nm, formed in base electrolyte, 1 wt% LiBF₄, and 1 wt% LDFOB electrolytes, respectively. It was notable that the SEI formed with LiBF₄ additive was much loose, while the SEI formed in base electrolyte and 1 wt% LDFOB electrolyte was compact. The SEI formed with LiBF₄ additive was more like loosely attached crystals rather than an integrated part of surface layer. The EDX data showed that the SEI formed with LiBF₄ contained the most of F. That was resulting from the LiF crystals produced by the decomposition of LiBF₄ [26]. For SEI formed in 1 wt% LDFOB electrolyte, the EDX data showed that the SEI contained more O and F than the SEI formed in base electrolyte. That suggested that LDFOB involved in the SEI formation.

Fig. 2 TEM images of a pristine graphite and graphite cycled three times in **b** base electrolyte, **c** 1 wt% LiBF₄ electrolyte, and **d** 1 wt% LDFOB electrolyte



However, to investigate the further sources of effects on the storage life of the full cell, it was essential to analyze the characteristics of normal anode with and without additives. Slices of the normal anode extracted from the full cell were analyzed. Cells were disassembled in Ar-filled glove box. For EIS test, slice of anode with diameter of 6 mm was cut from the anode and three-electrode cell was set up. For SEM, TEM, STEM, XPS, and FTIR-ATR analysis, the anodes in full cells were purified by DMC before observation.

The anode EIS analysis

The EIS was an efficient way to analyze the SEI properties [27]. The Nyquist plots of three-electrode cell with anode, extracted from cells before and after storage at 55 °C for 150 days with and without additives, as working electrode, Li/Li⁺ as reference electrode, and Li foil as counter electrode were depicted in Fig. 3. The equivalent circuit was showed in Fig. 3a. The resistance R_1 , representing the resistance due to the electrolyte, was given by the intercept of impedance curves with the real axis in the Nyquist diagram. The semicircle, appearing at high frequency, represented the solid electrolyte interphase resistance of anode. The semicircle, appearing at low frequency, represented the charge-transfer resistance. The Warburg element W_0 represented the charge diffusion in the electrodes [27–29].

The fitting results of SEI resistance and charge-transfer resistance were shown in Table 2. It was clear that after storage aging, both of the SEI resistance and charge-transfer resistance would increase. The increasing of SEI resistance was attributed to the growing of SEI after storage aging. While higher charge-transfer resistance R_{ct} generally corresponded to lower kinetics of the faradic reaction on the SEI, related to resistance of the anode/electrolyte reaction during storage aging [29]. The anode with 1% LiBF₄ electrolyte had the biggest growing of SEI resistance and charge-transfer

Table 2 The simulated values R_{SEI} and R_{ct} before and after storage aging at 55 °C for 150 days in different electrolytes

	R_{SEI}	R_{ct}
Base electrolyte (fresh)	17.7	4.9
LiBF ₄ electrolyte (fresh)	21.5	17.6
LDFOB electrolyte (fresh)	18.7	7.0
Base electrolyte (aged)	40.1	18.8
LiBF ₄ electrolyte (aged)	103	136
LDFOB electrolyte (aged)	21.6	27.0

resistance. The excessive increase of the resistances indicated that the SEI was significantly increased. Meanwhile, the SEI resistance of anode with 1% LDFOB was basically unchanged after storage aging and the charge-transfer resistance was significantly increased. It indicated the addition of LDFOB efficiently restrained the growing of SEI and anode/electrolyte reaction during storage aging.

Surface analysis

SEM was performed to investigate the surface morphology of the normal anode before and after aging at 55 °C with and without additives, depicted in Fig. 4. On the pristine anode, as seen from Fig. 4a, the carbon black particles with a diameter of 100 nm scattered on each of the graphite particle. After the anode charge-discharged in different electrolytes, we could see that there is no significant difference in the surface morphology between the pristine anode (Fig. 4a) and the anode activated in base electrolyte (Fig. 4b) or LDFOB electrolyte (Fig. 4f) since the even-coated SEI formed on anode was not thick enough to be observed from SEM. But on the surface of anode activated in the 1 wt% LiBF₄ electrolyte (Fig. 4d), a loose layer was

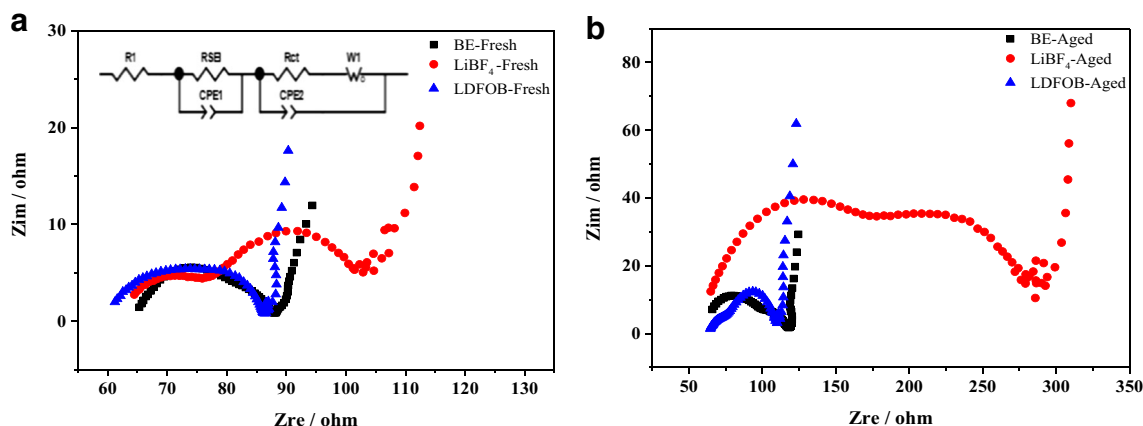
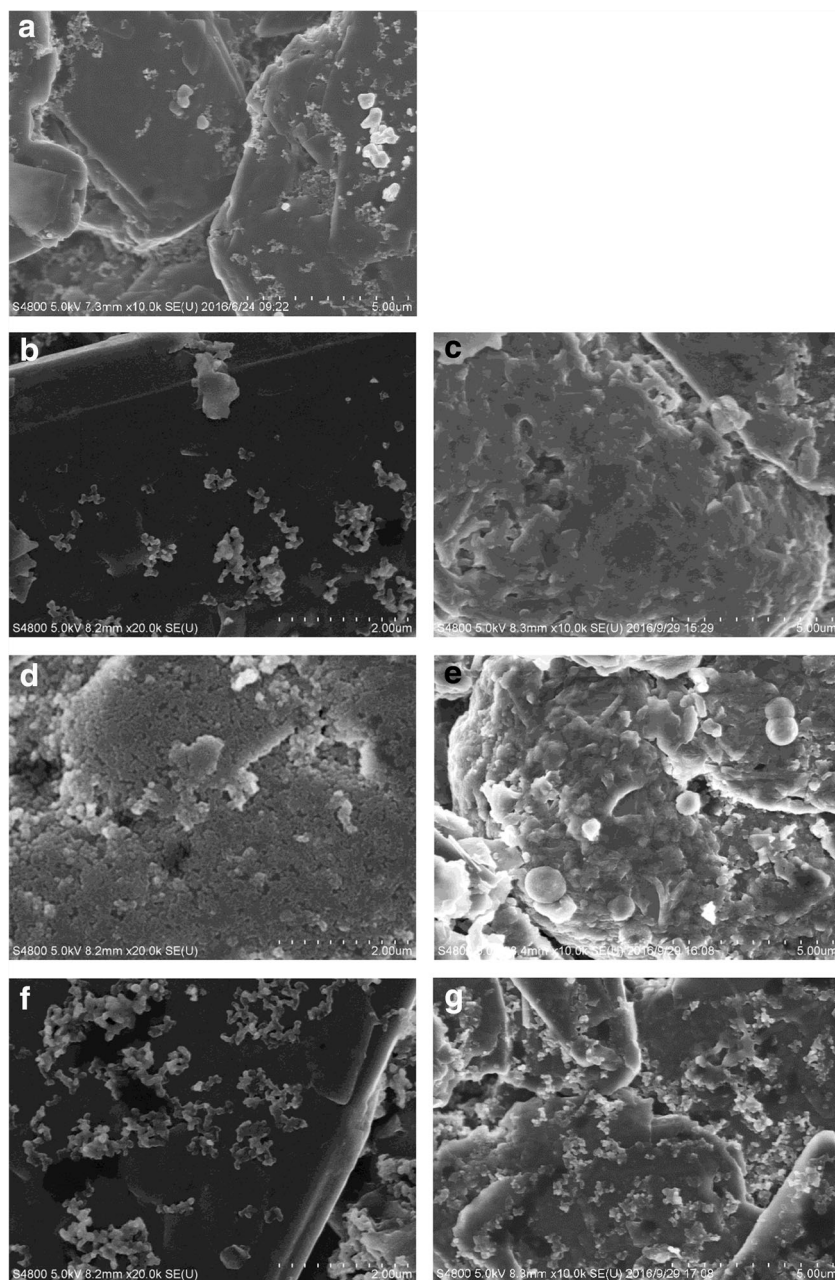


Fig. 3 Nyquist plot of the three-electrode cell with anode extracted from cells with different electrolytes as working electrode: **a** cells before storage aged and **b** cells storage aged at 55 °C for 150 days

Fig. 4 SEM image of anode (**a**: pristine anode; **b, d, f**: anode with base electrolyte, 1 wt% LiBF₄ electrolyte, and 1 wt% LDFOB electrolyte before storage aging; **c, e, g**: anode with base electrolyte, 1 wt% LiBF₄ electrolyte, and 1 wt% LDFOB electrolyte after storage aging at 55 °C for 150 days)



generated and coated on the graphite particle. As discussed above, the surface morphology of anode before storage, observed by SEM, showed the same characteristic of surface morphology of pure graphite anode observed by TEM. After aging at 55 °C for 150 days, all anodes were covered by a significant layer due to the SEI growing as showed in Fig. 4c, e, g. The layer on the anode with 1 wt% LiBF₄ electrolyte was much rougher and thicker than the others, and it was much more compact than its initial SEI (Fig. 4d), resulting in increase of impedance as showed before. Meanwhile, the layer on anode with 1 wt% LDFOB electrolyte was thinner since the carbon black

particle could still be observed. This suggested that the addition of 1 wt% LDFOB efficiently restrained the SEI growing.

TEM and STEM were carried out to get a deeper observation of the aged anode SEI after storage aging. Figure 5 depicts a typical TEM image of graphite particle after aging at 55 °C for 150 days in base electrolyte. As shown in Fig. 5, a thick layer was generated and fully coated on the graphite particles. But it was difficult to judge the thickness or the edges of the SEI from TEM observation since the SEI became too thick. To get a better view of the aged SEI, the STEM observation was carried out as Fig. 6

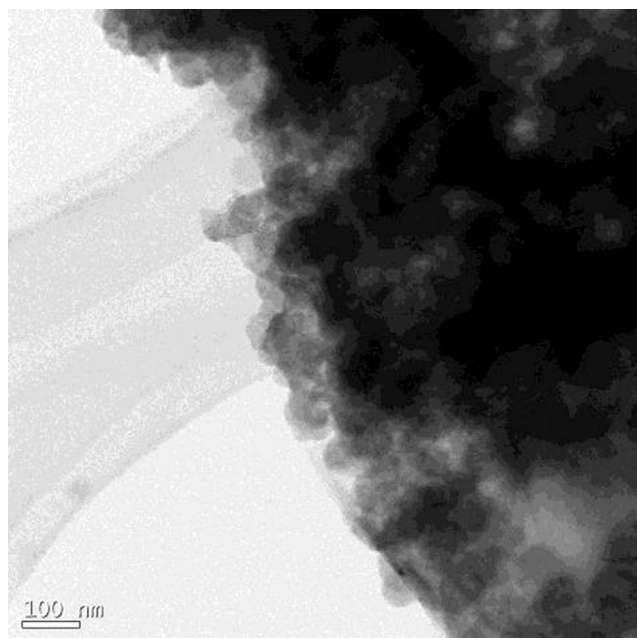


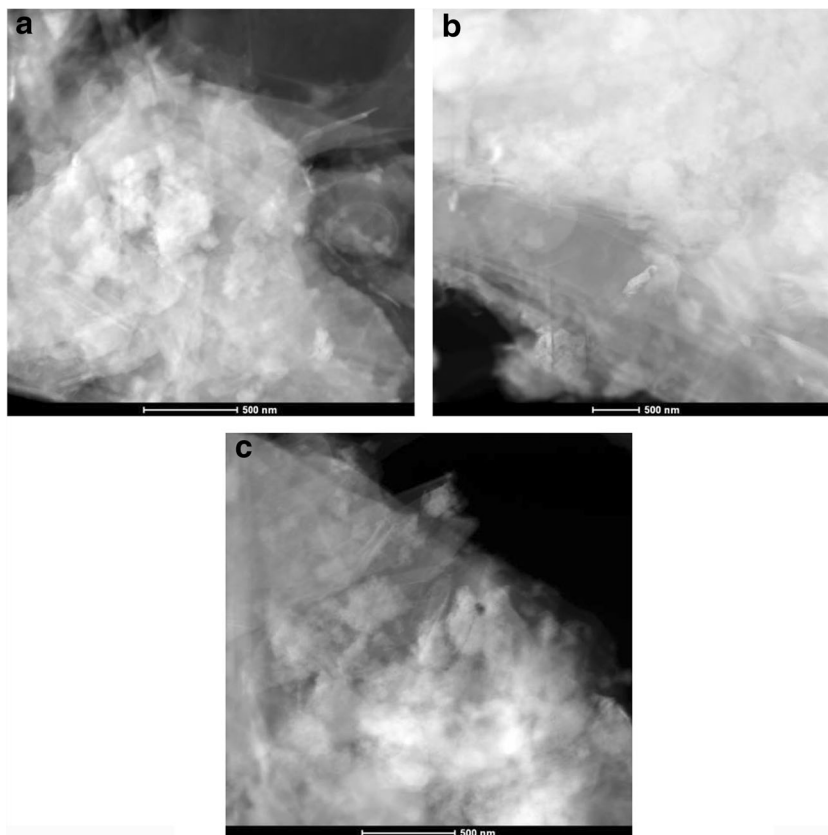
Fig. 5 A typical TEM image of graphite particle after aging at 55 °C for 150 days in base electrolyte

showed. The surface morphology observation showed in STEM displayed the same characteristics with the SEM observation: The surface of graphite articles, aging with

base electrolyte, was covered by a significant and compact SEI layer. On the surface of graphite aged with 1 wt% LiBF_4 electrolyte, the aged SEI was rougher and thicker, while the aged SEI in 1 wt% LDFOB electrolyte was much looser and thinner. This reconfirmed that the addition of LDFOB restrained the SEI growing while the SEI growing with the addition of LiBF_4 was promoted.

In order to get a better understanding of the changes on the anode surface after storage aging at elevated temperature, electrodes were analyzed by XPS before and after storage aging at 55 °C with and without additives. The $\text{C}1s$, $\text{F}1s$, and $\text{P}2p$ core peaks were shown in Fig. 7. Since the depth of XPS observation was about 5 nm, the XPS spectra exhibited the element components of top surface of SEI which was newly produced during storage aging. The analysis of surface atomic concentrations of electrodes before and after aging was shown in Table 3. Furthermore, the $\text{O}1s$ and $\text{Li}1s$ core peaks were not presented as it provided very poor information, since the O- and Li-containing species were with few variations in binding energy. Also, the $\text{B}1s$ core peak was not analyzed since the $\text{P}1s$ core peak also presented in the binding energy region (about 188~196 eV) and the peak appeared between 188~196 eV, observed from the anode with or without additives, was almost the same with the $\text{P}2p$ peak.

Fig. 6 STEM images of the graphite after storage aging at 55 °C for 150 days in **a** base electrolyte, **b** 1 wt% LiBF_4 electrolyte, and **c** 1 wt% LDFOB electrolyte



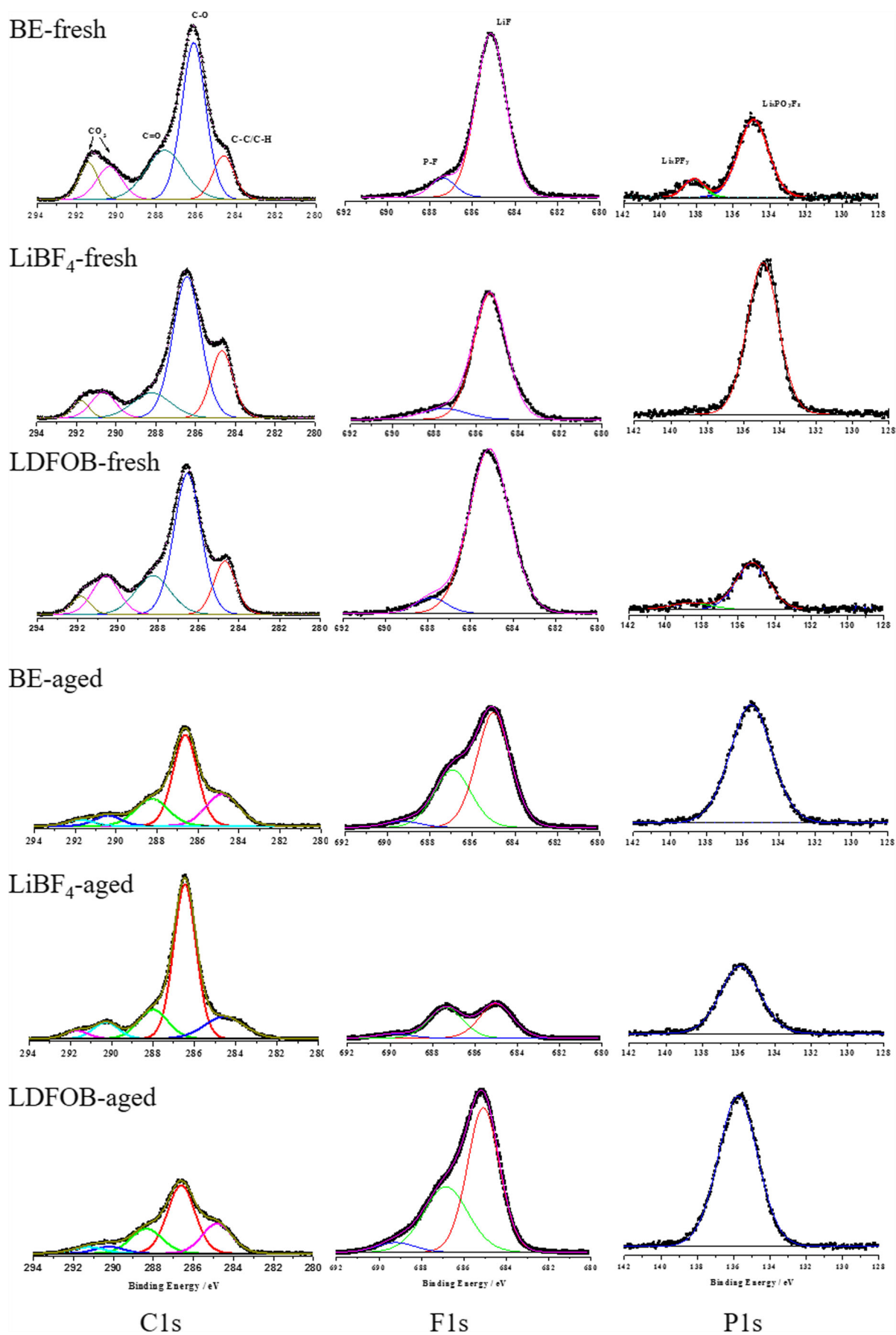


Fig. 7 XPS spectra of the graphite anode before and after storage aging at 55 °C for 150 days in base electrolyte, 1 wt% LiBF₄ electrolyte, and 1 wt% LDFOB electrolyte

In the C1s spectra, the peak observed at 284.5 eV was assigned to hydrocarbon contaminations and C atoms bound only to C or H atoms. The second, at 286.5 eV, was attributed to C–O bond which was originated from ethers and carbonates. The peaks, observed at about 288.2 eV, was attributed to C=O bond. The last two adjacent peaks, at about 290.5–291.5 eV, could be Li₂CO₃ and lithium alkyl carbonates [18, 30–34]. In fresh anode, as seen from the C1s spectra, the three peaks at 288–291.5 eV were less pronounced with addition of 1 wt% LiBF₄. The atomic concentration analysis in Table 3 showed that the content of the three peaks of the anode with 1 wt% LiBF₄ was about 50% of the anode with BE or LDFOB electrolyte. Meanwhile, the peak at 284.5 eV was significantly increased. This might result from the fact that the SEI was not even formed and coated on graphite with the addition of LiBF₄, as showed in TEM before. Also, with the addition of LDFOB, the peak at 291.5 eV was less pronounced than the base electrolyte while the other peaks were almost the same, this could result from the oxalate in LDFOB which involved in the anode SEI formation, restraining the decomposition of electrolyte solvent and the production of carbonates. After storage aging at 55 °C for 150 days, for 1 wt% LiBF₄ electrolyte, the peak at 286.5 eV was increased and the other peaks were almost the same, while all peaks after 284.5 eV were weakened in

base electrolyte. This suggested that the decomposition of electrolyte solvent was sustained in 1 wt% LiBF₄ electrolyte during aging. For base electrolyte and 1 wt% LDFOB electrolyte after aging, all peaks after 284.5 eV were attenuated and the elements concentration, represented by the peaks, was lower in 1 wt% LDFOB electrolyte than in the base electrolyte. This suggested that the decomposition of electrolyte solvent was restrained due to the SEI growing during aging and the 1 wt% LDFOB electrolyte was more effective on the restraints.

In the F1s spectra, the peak at 685 eV was attributed to LiF while the peaks at 687 and 689 eV were attributed to F–P which originated from Li_xPO_yF_z and LiP_xF_y, decomposition products of LiPF₆ [24, 31, 33, 34]. As seen from the F1s spectra, the main F-containing product was LiF in fresh anodes with different electrolytes. While after storage aging at 55 °C for 150 days, the atomic concentration of LiF and P–F was both increased in BE and 1 wt% LDFOB electrolyte. The increasing total content of F and decreasing of content of C proved that, after storage aging, the decomposition of electrolyte solvent was restrained due to the protective effect of growing SEI while the decomposition of LiPF₆ lasted in base electrolyte and 1 wt% LDFOB electrolyte. Meanwhile, in 1 wt% LiBF₄ electrolyte, the content of LiF was decreased while the concentrations of P–F were increased, but it had the least changes in the total F content. This indicated that the decomposition

Table 3 The surface atomic concentration of the fresh and aged anode with and without additives

	Binding energy	Atomic concentration					
		Fresh anode			Aged anode		
		BE	LiBF ₄	LDFOB	BE	LiBF ₄	LDFOB
C1s	284.3~284.8	2.31	5.43	4.56	4.69	4.33	3.88
	286.4~286.8	14.13	14.3	13.76	9.94	17.06	8.23
	288.2~288.7	6.29	3.47	6.09	4.00	4.24	3.41
	290.7~291.3	3.14	2.31	3.73	1.24	2.04	0.85
	291.5~291.8	3.45	1.05	1.29	0.73	0.93	0.91
F1s	685.1~685.3	7.25	6.3	9.77	12.51	4.19	14.01
	687~687.3	1.34	0.24	0.53	6.91	3.83	8.83
	689.2~689.9				0.71	0.55	1.1
P2p	135~135.5	1.02	2.00	0.65	3.16	1.96	3.57
	138.6~138.7	0.21		0.11			
O1s		28.96	28.42	25.97	18.00	24.00	16.26
Li1s		31.00	34.81	32.26	37.21	36.5	38.02

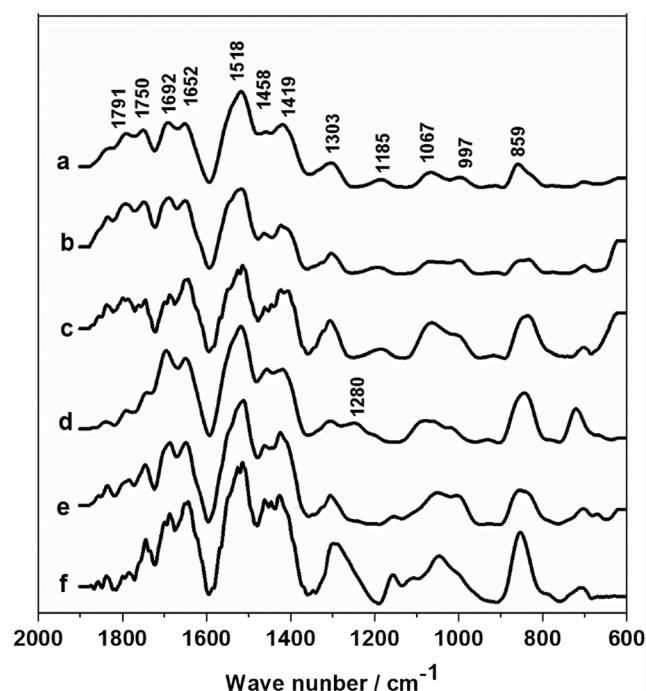


Fig. 8 FTIR-ATR spectra of anode extract from cell before (a, b, c) and after (d, e, f) storage aging with base electrolyte, 1 wt% LiBF₄ electrolyte, and 1 wt% LDFOB electrolyte, respectively

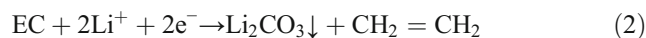
of LiPF_6 and electrolyte solvents were both lost during the storage aging.

In the $\text{P}2p$ spectra, the peak at 135.2 eV was attributed to the P–O bond which originated from $\text{Li}_x\text{PO}_y\text{F}_z$, while the peak at 138.5 eV was attributed to the P–F bond which originated from LiP_xF_y [31, 33, 34]. As seen in the spectra, the content of LiP_xF_y was low in the fresh and aged anode. The peak at 135.2 eV was increased in base electrolyte and 1 wt% LDFOB electrolyte but basically unchanged in 1 wt% LiBF_4 electrolyte, exhibiting the same changes as the peak at 687 eV in $\text{F}1s$ spectra after storage aging with different electrolytes since both peaks were representing the $\text{Li}_x\text{PO}_y\text{F}_z$ component.

Analysis of the graphite anode by FTIR-ATR spectra provided additional insight into the differences of electrode surface after storage aging at elevated temperature with different additives. As shown in Fig. 8, the IR spectra of anode with different electrolytes were all similar. The well-defined absorption bands, located at 1791, 1652, 1510, 1419, 1303, 1067, and 859 cm^{-1} , could be attributed to the solvent reduction products ROCO_2Li and Li_2CO_3 [18, 35, 36]. The peak at 1185 cm^{-1} (P–F) and 1280 cm^{-1} (P–O), attributed to the $\text{Li}_x\text{PO}_y\text{F}_z$ [18, 35, 36], became stronger after storage aging in base electrolyte and LDFOB electrolyte, indicated the content increase of $\text{Li}_x\text{PO}_y\text{F}_z$ after storage aging. The peaks at 1303, 1067, and 997 cm^{-1} , representing the C=O and C–O, respectively [36, 37], were much stronger with the addition of LDFOB in fresh and aged anode. This suggested that the oxalate in LDFOB involved in the anode SEI forming and aging, in coordination with the data in XPS analysis.

Storage aging mechanism discussion

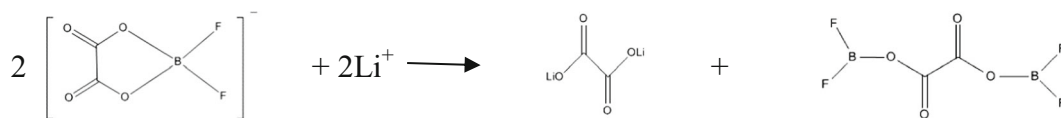
The Li^+ and e^- loss in anode and the SEI growing were the main causes of storage aging. The main SEI components, including ROCO_2Li , Li_2CO_3 , LiF , and $\text{Li}_x\text{PF}_y\text{O}_z$, were originated from the following reactions (EC as an example for the electrolyte solvent) [25, 38]:



The reactions 1–4 would last from the SEI formation to the SEI aging. It was obvious that the decomposition of electrolyte solvent would directly consume Li^+ and e^- and lead capacity loss during storage aging, while reactions 3–4 would not directly induce Li^+ and e^- loss in anode. The SEI on anode acted as a protective layer that inhibited the transportation of Li^+ and e^- and restrained the aging reaction 1–2 but it had less impacts on the aging reactions 3–4. Hence, after storage aging, the newly produced SEI would contain less carbonates (Li_2CO_3 and $(\text{CH}_2\text{OCO}_2\text{Li})_2$) but more decomposition components of LiPF_6 (LiF and $\text{Li}_x\text{PF}_y\text{O}_z$) due to the protective effect of SEI, which had been confirmed in the atomic concentration analysis via XPS showed before.

The initial SEI formed with addition of LiBF_4 was as thick as about 100 nm but the SEI resistance was basically the same as base electrolyte. That was attributed to the generation of the LiF crystals, produced by the decomposition of LiBF_4 , scattering on the graphite particle and forming a loose SEI layer. The thick but loose SEI, as seen in the TEM and SEM observations before, could not efficiently protect the electrolyte solvent from the decomposition with Li^+ and e^- on the graphite SEI. After storage aging, the components of the newly produced SEI were basically the same with the initial SEI as XPS showed, made the total SEI thick and compact. But it consumed much Li^+ and e^- in anode and led significant capacity loss.

The initial SEI formed with the addition of LDFOB was about 20 nm and even-coated on the graphite surface. It was thicker than the SEI formed in base electrolyte (~10 nm). The surface analysis (EDX, XPS, and FTIR-ATR) indicated that the oxalate in LDFOB involved in the SEI formation. The probable products could be $\text{Li}_2\text{C}_2\text{O}_4$, produced by the following reactions [18, 39]:



The thick and compact SEI could efficiently protect the Li^+ and e^- loss in graphite, restraining the decomposition of the electrolyte solvent. Element analysis revealed that the newly

produced SEI contained less carbonates after storage aging which reconfirmed that decomposition of the electrolyte solvent was restrained.

Conclusions

The impacts of LiBF_4 and LDFOB as additives on the storage performance of $\text{LiNi}_{0.8}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2/\text{graphite}$ cells in full state of charge at 55 °C were investigated. After storage at 55 °C for 150 days at full state of charge, the capacity retention was 75, 18, and 80%, for base electrolyte, 1 wt% LiBF_4 electrolyte, and 1 wt% LDFOB electrolyte, respectively. The estimate storage lifespan of the cell at 55 °C was about 90, 30, and 150 days, respectively. The ex situ observation of the anode surface characteristics revealed that the SEI formed in 1 wt% LiBF_4 electrolyte was about 100 nm, resulting from the LiF crystal produced by the addition of LiBF_4 . But the thick SEI was a loose layer and could not prevent the decomposition of electrolyte on anode during storage aging. The SEI formed in 1 wt% LDFOB electrolyte was about 20 nm, resulting from the involvement of lithium oxalate in SEI formation. The compact SEI was more effective in preventing the Li^+ and e^- loss in anode and the decomposition of the electrolyte solvent than the SEI formed in base electrolyte, resulting in a longer storage life.

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